

VolTree

OHLCV (stock i)
 $\rightarrow \text{pk}_{i,t}$

earnings calendar
(external)

endogenous $\mathbb{R}^8$
HAR lags + momentum

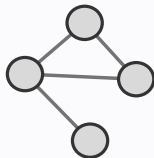
earnings $\mathbb{R}^4$
distance to release

Gamma-loss XGBoost
300 trees · 31 leaves · reg:gamma (= QLIKE)

per-stock forecast $\hat{\sigma}_{i,t+h}^2$
 $h \in \{1, 5, 10, 22\}$

$\hat{y}_i$ (all stocks)

leaf-cooccurrence graph smoothing



$$\hat{y}_i^{sm} = (1 - \alpha) \hat{y}_i + \alpha \bar{\hat{y}}_{\text{kNN}(i)}$$

α fit on validation

kNN on shared leaves ($k = 10$)